From a failed run to a controlled replay of the step where it failed

1 Trace

compare each failed output with the reference; find the smallest change that removes the error

2 Locate

read the transcript back to the step where that change was due; note what the agent had printed and read

3 Capture

take the exact model request sent right after the disproving output: system prompt, tools, every message

4 Change one thing

append one note, or change the effort, and sample the next turn 6 times with Claude Haiku 5.5

5 Score

a judge model reads the turn's text and code, never the condition, against a criterion fixed first

104 failed outputs → 13 failure points, plus 3 points from passing runs as controls → 12 conditions × 6 samples per point

Steps 1 and 2 are the tracing (Section 3.3); steps 3 to 5 are the replay (Section 3.4).